DECEPTIVE SOUNDS: AUDITORY ILLUSIONS & ARTEFACTS IN SOUND ART

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ABSTRACT

There is a rich history in the visual arts of works which use optical illusions that confuse and excite the eye. In Sound Art, there hasn't been established movements which have focused on auditory illusions. However, there are a range of works which share similar experiential themes. This paper examines the artefacts and oddities of our aural sense, and catalogues how these interesting perceptual phenomena have been explored in sound art. This cataloguing will connect the implicit experiential and perceptual themes that are at force in a range of works, and will speculate on underexplored fields for sonic exploration. To this purpose, two new sound works by the author, *Destructive Passages One & Two* and *rise.risset*, are discussed.

1. INTRODUCTION

Optical illusions have been a source for many artistic works and movements. The realism of the Trompe-l'œil movement, the impossible forms of M.C. Esher, and the paradoxical movement in the static images of the Op Art practitioners all offer interesting new perceptual confusions. Through confusing and tricking the senses, they invite the audience to meditate on their perceptual apparatus. In experience of these works, we can become aware of the subjectivity and deficiencies of our perceptual apparatus, and its constant mediation of our world.

The eye and the ear have their own distinct artefacts and oddities, which can be explored though different forms of illusions. In sound art, there hasn't been established movements which have focused on auditory illusions. However, there are a range of works which share similar experiential themes. This paper examines the different artefacts and oddities between our visual and aural senses, and catalogues how these have been explored in sound art.

The field is organized into different forms of effects. Impossible Sounds collect two auditory illusions that are perceived to have perpetual motion: Shepard tones and Risset rhythms. They share commonalities with works by M. C. Esher and other artists, who used impossible objects in their works. The next category is a large collection of non-linear perceptual effects. These are experiences where the auditory system adds distortions and artefacts to our experience. Lastly, the Phantom Sources section covers a range of spatialisation techniques which have a rich history in electroacoustic music.

Through organising the modes of perception framing that artists have employed in their works, some of the differences between our visual and aural senses will be teased out. This cataloguing will connect the implicit experiential and perceptual themes that are at force in a range of works across multiple mediums, and will speculate on underexplored fields for sonic exploration. To this purpose, two new sound works by the author, *Destructive Passages One & Two* and *rise.risset*, are discussed before some final discussion about future works.

2. IMPOSSIBLE SOUNDS

Impossible sounds are auditory illusions which imply an impossible sonic behaviour. Shepard tones create the effect of perpetually rising or falling pitch motions. This idea was then translated to the temporal realm. Risset rhythms sound as if they are perpetually accelerating or slowing down.

2.1. Shepard Tones

The use of Shepard tones has become prevalent in a range of different sonic expressions and mediums. The Shepard tone, named after Roger Shepard, is an illusion of perpetual pitch motion (Shepard, 1964). The illusion creates a set of musical notes or tones in which the pitch relation between the notes is perceived as cyclical rather than linear, disrupting our normal sense of low and high. The tone is perceived as eternally rising or falling by using multiple octaves of tones superimposed over each other. The amplitude of these tones is controlled to keep a constant focus on the middle of spectrum with a stationary Gaussian envelope (Vernooij 2016). As the pitch of the tone moves further away from the centre, it decreases in volume, and as it enters the perceptive field from below, it slowly increases in volume. This is carefully controlled

so that at the end of a complete octave cycle, the sound is spectrally identical as at the beginning, creating an eternal perceptual cycle.

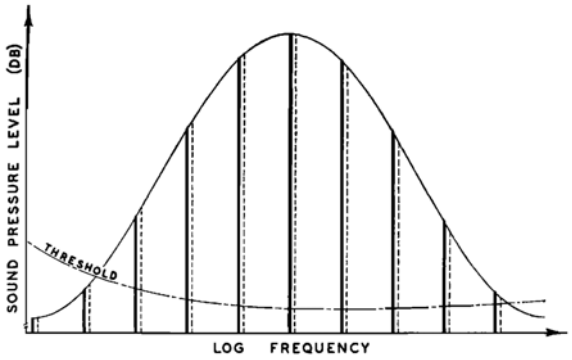


Figure 1. Spectral envelope of a Shepard tone, with the dotted lines showing the direction of a Shepard scale.

Using the new audio technology of the 1960's, Jean-Claude Risset expanded the potential of this illusion by developing a variant of Shepard's discrete scale for continuous gliding tones, now known as the Shepard-Risset Glissando (Risset, 1986). The computer generated synthesised tones opened up the possibilities of the illusion, and has become the more common presentation of the Shepard tone.

The experience of the Shepard tone has been likened to this impossible circularity of the Penrose stairs (Penrose and Penrose, 1958). The stairs share this impossibility of eternal cyclic motions, with four 90-degree turns which form a continuous loop. This means that a person could climb them but never get higher or lower. This impossible architecture was influenced by, and influential to, M. C. Esher who in 1960 created the lithograph print *Ascending and Descending*. The impossible form sits at the top of a building with people walking up and down the never-ending staircases.

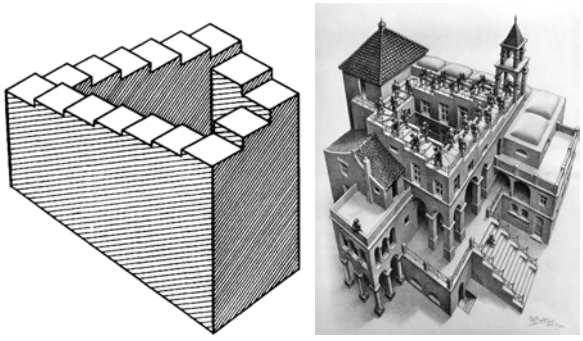


Figure 2. *Left* - Penrose stairs, *Right – Ascending and Descending* – M. C. Escher.

The use of Shepard tones and the Shepard-Risset Glissando has become widespread with some scholars arguing that its musical roots can be heard in the pitch circularity of pieces written as far back as 1550. These works show similarity in their use of a constant upwards or downwards pitch trajectory, which is negated by octave doubling to return to an initial point. Perceptually, this circularity is not as illusory as the Shepard tone, as it's much easier to trace the individual movement of the layers.

The affective element of the Shepard tone makes it a versatile sound design tool in films. Film composer Hans Zimmer and Director Christopher Nolan, have often used the illusion in their films to create tension and suspense. In *Dunkirk* (2017), the eternal upward motion supports the increasing tension throughout the movie of the three intertwined story lines. In *Inception (2010)*, the illusion supports the themes of surrealism and questioning of reality that lies at the heart of the film. The Batpod in *The Dark Knight Rises* (2008), uses a Shepard tone for its engine noise to sound as if it were continuously increasing in speed (Jackson, 2008).

The illusion has also been used in popular music, with post-rock band Godspeed You! Black Emperor's album *F♯ A♯ ∞* using a Shepard tone as an opener, and Pink Floyd's song *Echoes* concluding with one. Austrian composer Georg Friedrich Haas utilizes the effect in his large ensemble piece *In Vain (2000).*

In another context, the Shepard tone has been used to show another peculiarity of our pitch perception. Diana Deutsch conducted a perceptual psychology experiment in which participants listened to two Shepard tones played sequentially and spaced a tritone apart (Deutsch, 1986). In each case, they reported whether they heard the two tone sequence as ascending or descending. Many striking observations were drawn from this experiment. The participants pitch perception wasn't absolute, but more relative, with the pitches being organized in a circle. Consistently, participants perceived tones in one region of the pitch circle as higher, and those in the opposite region as lower. Which region was ascending and descending differed largely between participants. Although the tritone paradox doesn't feature in many artworks or music, it does unveil one of the subjectivities of our cognitive processing of sound. The wide range of responses is idiomatic of many of these illusions.

2.2. Risset Rhythms

Jean-Claude Risset was instrumental in expanding the potential of the discrete Shepard scale through developing the continuous Shepard-Risset Glissando. Risset's research didn't stop there, as he created a similar illusion in the temporal domain, although he credited Kenneth Knowlton with being the first to synthesise the effect (Risset, 1986). Through an analogous method, Risset synthesised rhythmic phrases that are perceived as eternally accelerating or decelerating. Instead of multiple octaves being superimposed, multiple versions of the same rhythm are superimposed at different tempo divisions. The stationary Gaussian amplitude envelope is linked to tempo, and keeps the centre layer the loudest, with the outer layers being progressively attenuated.

Risset himself explored these paradoxical rhythmic motions in multiple works. *Contre Nature* (1996) for percussion and fixed media uses multiple rhythmic octaves of two times, four times and eight times as fast to create an eternally accelerating trajectory. Risset

expanded this idea to having musical sections that nested contradicting temporal trajectories. *Moments newtoniens* (1978), *Mirages* (1978), *Electron-Positron* (1989), all have rhythms that speed up constantly, yet are slower at the end than where they started at the beginning. Risset also explored the pitch equivalent to this in *Moments newtoniens*, and *Mutations*, where the pitch seems to ascend, yet eventually is lower in pitch than where it began. Other works which explore Risset rhythms include Autechre's *Fold4, Wrap5*, which uses extended periods of decelerating Risset rhythms.

The musical application of Risset rhythms haven't been as explored as their pitch counterparts. This may be for many reasons, with Dan Stowell suggesting that the complexities of managing accelerating fractal rhythms within compositional and sound-design environments may have held back their usages (Stowell, 2011). In an attempt to increase their accessibility and help facilitate their use in compositions, Stowell has derived mathematical expressions for scheduling accelerating Risset rhythms.

3. NON-LINEAR SOUNDS

The aural sense is a non-linear filter. This means that it doesn't possess the three main characteristics of a linear filter which are as follows. Firstly, the output of the filter must never contain any frequency component that was not present in the input signal. Secondly, if the amplitude of the input to the filter is changed by a certain factor, then the output should also change by the same factor. And lastly, if a number of sine wave inputs are applied to the filter simultaneously, then the resulting output should match the output that would be produced if the inputs were presented separately, and their individual outputs summed (Mather, 2009).

Non-linear sounds are a group of sounds which reveal the non-linearity of our auditory perceptual processing. They highlight how our hearing differs from a transparent linear filter, and the way it colours our experience.

3.1. Combination Tones

Combination tones are an illusory tone which are a by-product of the interaction of two real tones. There are two types: sum tones whose frequency is equal to the summation of the two real tones. And difference tones, whose frequency is equal to the difference between the two real tones. Combination tones have an important place in history, as they were one of the auditory phenomena that spurred investigations into the non-linearity of the ear. Their discovery is attributed to violinist-composer Giuseppe Tartini, who used the third tone in correcting faulty intonations of double stops on the violin (Jones, 1935). Through making the third illusory tone consonant with the two real tones, he was guided in finding the intonation of the open strings.

The musical uses of combination tones have been varied through electroacoustic and instrumental music. The consideration of the relationship between partials and sine tones makes combination tones well aligned with the Spectralist school. They form a driving melodic tool in Tristan Murail's *Ethers* (1978), with the purity of the solo flute being supported by different combinations of the ensemble. Claude Vivier, who was heavily influenced by the Spectralist school, explored combination tones as a timbral device, to produce a distinctive spectral harmony that became a core element of his compositional voice (Gilmore, 2014). Ben Hjertmann has made combination tones as harmonic materials the centre of his doctoral thesis (Hjertmann, 2013), producing a new work for small ensemble, *Angelswort* (2012), and vividly analysing Ezra Sims's *Quintet* (1987).

In the electroacoustic world, La Monte Young has extensively explored relationships between pure sine tones, and often created difference tones in his piano and installation works. *Dream* House (1995) creates a physical sound environment that offers the audience dynamic autonomy to experience and explore the different tones and their effects. Various wavelengths of static tones ricochet around the room, projected from loudspeakers, and blend to make a complex typology of pure tones and combination tones. Walking around the space, different tones blend at different spatial nodes, and even a slight movement of the head can create a different effect (Grimshaw, 2011).

3.2. Otoacoustic Emissions

Heavily related to combination tones, Otoacoustic Emissions (OAEs) are sounds which the ear makes itself. The emission is not an illusion, but instead an artefact of the physiology and non-linearity of the auditory system. OAEs are sounds that are generated from within the inner ear. This phenomenon is related to the amplification function of the outer hair cells in the cochlea (Mather, 2011).

Otoacoustic Emissions can be provoked through three different types of stimulus. Stimulus Frequency OAEs are created by a single pure tone, with the resulting tone matching the stimulus tone (Miller et. Al, 2004). Transient-evoked OAEs are stimulate by a loud click sound (Kemp, 2002). Lastly, Distortion product OAEs are created when a pair of closely spaced tones with a ratio between 1.1 and 1.3 are presented with high intensity (Hall, 2000). This form of OAE's has had the most musical exploration, as they create a different tone to the stimulus, which allows for the listener to be able to easily distinguish between the two pure tones and the third artefact tone.

Maryanne Amacher's seminal work with OAEs, *Head Rhythm/Plaything* from her album *Sound Characters (Making the Third Ear)* creates arpeggios of OAE's that move around the head in disorientating patterns. The localisation of these sounds creates an interesting contrast between the 'outer' sounds, and 'inner' sounds, creating a complex spatial counterpoint between real tones and generated tones. Amacher discussed this exploration of the spatial elements of the two types of tones as *perceptual geography*. She writes 'I distinguish *where* the tones originate, in the room, in the

ear, in the brain, in order to examine this map and to amplify it musically' (Amacher, 2004).

In a similar exploration of the spatial elements of OAEs, Alex Chechile has recently composed a series of pieces entitled *On the Sensations of Tone* (2010). The works mix electronic and acoustic sources, using OAEs to invite the listener into an exploration of their listening space. Through slight head movement, OAEs will appear, disappear, and change timbre. Macroscopic and microscopic listening can be achieved by shifting attention between the primary tones and the ear tones. The effect can be enhanced by cupping hands around one's ears.

In composing *On the Sensations of Tone*, Chechile has developed *The Ear Tone Toolbox*, a collection of Max externals, VST plugins, and patches for the hardware OWL synthesiser, that offer creative control over OAEs (Chechile, 2015). The toolbox allows the user to input various combinations of evoked distortion products and acoustic primary tones, which are then synthesised in the software. This contribution to the field will allow more composers access to the creation and exploration of OAEs in their music, and may inspire a range of new works.

Other composers have explored different elements of the experience of OAEs. Jacob Kierkegaard makes them the explicit subject of his 38-minute work *Labyrinthitis* (2007). Kirkegaard has presented this work both in stereo form, and as a 16-channel sound installation, with the speakers suspended in a contracting spiral, modelled on the shape of the cochlea. The piece begins with two pure tones that excite a singular OAE. This precise tone is then used as the next acoustic tone, starting a process of cascading interplay between acoustic and OAE tones.

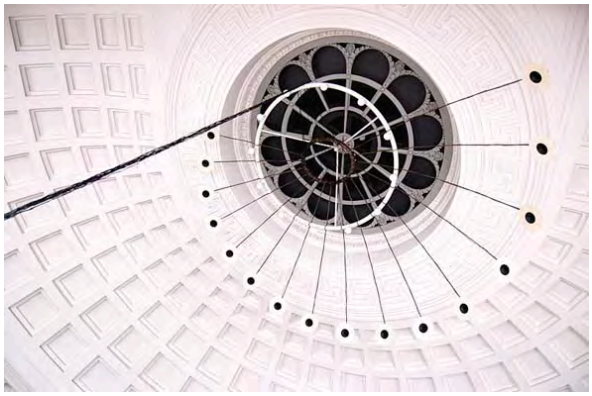


Figure 3. *Labyrinthitis* - Jacob Kirkegaard

One of the most intriguing elements of OAEs is on its effect on how we conceptualise the auditory system. Often, the ear is thought of as a passive, submissive sense organ. In comparison to our visual sense, our aural sense seems to be continuously engaged, always open to stimulus from all directions, and unable to be turned off. We can look away, but there is no sonic equivalent, and unlike our eyes, 'we have no ear lids' (Schafer, 1977). OAEs turns the ear into a dynamic element in the system, both receiver and producer, and problematizes this conception of it as a passive receptor.

3.3. Zwicker tone

Similar to the transient-evoked OAEs, the Zwicker tone is an auditory illusion produced in response to noise. In the case of the Zwicker tone, broadband noise with a spectral gap in the middle is presented for several seconds. The auditory illusion appears once the noise is removed, and a pure tone that falls within spectral gap is audible for up to six seconds. This is an interesting case of non-linearity, as the Zwicker tone demonstrates the auditory system hearing a sound which is not in the stimulus.

The Zwicker tone has not, to the authors knowledge, been explored in sound art yet. There may be multiple driving factors for this. The illusion needs extended exposure to loud noise to be provoked, which may cause discomfort for listeners. It is also a subtle experience, which may be hard to detect the nature of the illusion without previous knowledge.

However, this does not mean the Zwicker tone does not have potential, as it shares qualities with other sensory illusions which have been explored in art. A notable commonality is with the visual afterimage. This phenomenon forms an image that continues to appear in one's vision after the exposure to the original image has ceased. The nature of the visual afterimage ranges depending on the stimulus, including dark spots, illusory motion and changes in colour (Bayne et al., 2014).



Figure 4. *Your Blue Afterimage Exposed* – Olafur Eliasson

The colour afterimage forms the perceptual material of Olafur Eliasson's *Your Blue Afterimage Exposed* (2000) shown in Figure 4. An intense square of colour is projected onto the wall, flooding the room in a vivid orange. At timed intervals, the projector is turned off, suddenly depriving the visual senses of this immersive light. The complementary blue colour appears in the visual perception of the audience, creating a square that matches the orange one that has just vanished. The move from outside stimulus, to internal perception is heightened by the fact that now, when the audience looks away, the

blue square follows their gaze, revealing itself as part of their subjective perceptual processing.

3.4. Missing Fundamentals

Combination tones and Otoacoustic Emissions aren't the only sounds that provoke our cognitive processing to create new notes. Complex harmonic sounds are made up of a fundamental frequency, and a series of evenly spaced partials. The missing fundamental illusion arises when a set of partials are presented that lack the fundamental frequency. The listener will still perceive the lower fundamental, even though it is not present. This reveals an element of the way we perceive pitch. The brain can infer the fundamental pitch from the relationship between the partials (Cariani & Delgutte, 1996; Schwartz & Purves, 2004).

The experience of the missing fundamental is surprisingly common and will most of the time go unnoticed. Guitars, violins, and timpani all produce the illusion which helps to create the low perceived frequencies than their physical resonant bodies can support (Howard & Angus, 2006).

The missing fundamental illusion is also present in the technological reproduction of sound. Telephone conversations create the illusion, as the small speakers in telephone headsets can only generate frequencies in the range of 300 Hz to 3 kHz. Male voices can have fundamental frequencies of 150 Hz, and these low frequencies are still perceived because of the illusion (Mather, 2011). The effect has been utilized in a range of commercial D.S.P plugins which seek to allow speakers to seemingly produce loud bass frequencies that are lower than their technological limits. Through using the effect, the loud bass frequencies are implied by creating precise partials, which means that the dynamic range isn't consumed by large amounts of low frequency energy (Grimm et al, 2001).

Due to the missing fundamental effect usually going unnoticed, it hasn't been thoroughly explored in musical works. An exception again comes from La Monte Young, through his exploration of different tuning schemes in his highly influential work, *The Well Tuned Piano* (1964). Through retuning the strings to a seven-limit just intonation scheme, the combination of notes can create the effect of implying an unplayed fundamental (Gann, 1993). John Schaefer describes experiencing this passage as 'keyboard filigrees and shimmering harmonics …a strongly implied drone, although since it's several octaves below the range of the piano, it's never actually heard' (Schaefer, 1987).

3.5. Binaural Beating

Another form of non-linearity in our auditory system is displayed by the binaural beating effect. Similar to combination tones, binaural beating is caused by the relationship between two pure tones. In this case, the two pure tones must be below 1500 Hz and have less than 40 Hz difference (Licklider et al. 1950). When these tones are presented dichotically - one tone in each ear - a third tone is perceived. This third tone is not perceived like a normal pitch, instead it is often heard as a beating effect, amplitude modulating the other two tones and creating an illusion of motion in the sound. The interaction between the processing of these two sounds creates a confusion in localisation. As Gerald Oster describes it, 'Listening to binaural beats produces the illusion that the sounds are located somewhere within the head (Oster, 1973).

This illusory localisation unveils one of the mechanisms by which we are able to sense the direction of sounds. Low-frequency signals are localized primarily by detecting the difference in phase – the between the sounds reaching the two ears (Oster, 1973). At higher frequencies, the size of the head is larger than the length of the soundwaves, so the difference in phase becomes obscured. At these frequencies, we rely more on the interaural level and time differences to sense the location of sounds (Wallach, 1940). This change in method of detecting location explains why binaural beats must be below 1500 Hz, as it is the confusion in phrase difference in which the illusion manifests.

Many of the musical explorations of binaural beats have been through their supposed potential for brain entrainment. This is the potential for the listener to synchronize the electrical activity in ensembles of cortical neurons in their brain to an external stimulus (Will and Berg, 2007). The scientific testing of this has been problematic, and some researchers have found 'no effect of binaural beat frequency eliciting a frequency following effect in the EEG' (Vernon, 2014). Regardless, many websites and services have been created which devote themselves to binaural beats, with long pieces claiming to awaken intuition, universal healing, and open the third eye.

Binaural beats have a great potential for sonic exploration as they create a unique experience of sound seemingly emanating from within the listeners head. This play with spatialisation has a rich history in electroacoustic music, as will be discussed in the next section, Phantom Sources.

4. PHANTOM SOURCES

One of the most commonly used auditory illusions in sound art is spatialisation of sound sources. Spatialisation of sound has been a core aesthetic element in the development of electroacoustic music. A range of linguistic tools for describing space have been theorized, and many new technologies have been created. These tools seek to afford the ability to explore space as an expressive parameter. A full discussion of this large field is outside the scope of this paper, however, many of the technological advances are informed by the cognitive processes involved in locating the source of sounds.

The spatial position of sound is not represented directly in our auditory pathway (Oldfield, 1984). Therefore, spatial cues must be derived by the nervous system from different relationships within the sonic stimulus. The exact nature of these cues and the way spatial information is derived from them, has been the subject of research for many decades. This research has led to a thorough understanding of our capacity to

determine the location of a sound, from the sound alone. Binaural cues, which stem from differences in stimulus between the two ears, provide the major cues to localising sound on the horizontal plane. Two major binaural cues have been identified. These are differences of time (either onset differences or phase differences, depending on the frequency) and differences of intensity (Mather, 2011). However, with these binaural cues, there are multiple locations at which the sound could be originating from. For any one binaural cue there is a range of potential positions which could account for the same pattern of interaural differences.

Additional cues are derived from the spectral modifications that the external ear has on sound. The pinna act as a sound filter, selectively modifying the amplitude and phase of frequencies above 6 kHz. The asymmetrical shape of the ear means that the modification of sounds depends on the sound direction (Watkins, 1978).

Through understanding of the processes involved with auditory localisation, audio technologists and composers have been able to use and create new tools that make space a dynamic, compositional parameter. Through precise control of amplitude and filtering between multiple loudspeakers, sounds appear to be created from locations between and beyond the loudspeakers. This is not limited to stereo, with complex algorithms like Vector-Based Amplitude Panning and Wavefield Synthesis expanding this to the full performance space with large arrays of loudspeakers. These techniques also expand to three-dimensional space making elevation a compositional parameter.

5. NEW WORKS

Through investigating auditory illusions and artefacts as an artistic material, the author has created two new works: *Destructive Passages One & Two*, and *rise.risset*. Both works seek to offer a rich perceptual experience for the audience. They invite the audience to meditate on the subjectivity and uniqueness of their own perceptual experience through an interplay between illusion and simple materials.

5.1. Destructive Passages One & Two

Destructive Passage One & Two is a sound installation that explores the multiple artefacts in our perceptual processing of two pure tones. Through the two pairs of tuning forks that are slightly detuned from each other, the work provokes binaural beating and combination tones. Solenoids strike each fork, with a bamboo resonating cavity channelling the sound to a small area of space.

There are two different arrangements. The first channels the sound together, colliding the sound waves into each other, creating destructive and constructive interference. The difference between the two tones is 3 Hz, which manifests itself as amplitude modulation. Due to the small 3 Hz difference, instead of hearing the third illusory tone directly, we hear its effect on the other two tones. It creates a slow beating pattern, as the volume swells in and out three times a second. Due to the small space at which the two sounds are channelled into each other, this arrangement invites the audience to listen in with one ear. As they move their head, the blend between the two tones changes, controlling the intensity of the illusion effect.



Figure 5. *Destructive Passages One & Two* – Blake Johnston

The second arrangement invites the audience to listen to the two tones dichotically. The two resonating cavities point straight forward, created a gap slightly larger than the average head. The smaller 1.5 Hz difference between the tuning forks in this arrangement creates binaural beats. The audience can move their head, blending the two tones together, creating the illusory localisation of the tones moving around the head.



Figure 6. *Destructive Passages One & Two* – Blake Johnston. *Right* – Arrangement One. *Left* – Arrangement Two.

Both arrangements give the audience the autonomy of experience the work on their own terms. A button is pressed by the listener to actuate the solenoids, striking the tuning forks into motion. The illusions are contained to a node in space, allowing the audience to be free to move their head and investigate the space, changing the blend and intensities of the effects. This affordance of autonomy reinforces the demystification of the effect, as the audience is in control of many of the elements. They can see that what they are experiencing originates from the two simple tuning forks in front of them.

5.2. rise.risset

rise.risset is work for Bridget Johnson's *speaker.motion* (Johnson, 2017), a quartet of mechatronic speakers that can rotate through space to change the directionality of the speaker. The speakers can be controlled in real-time by MIDI messages, and can rotate and tilt simultaneously, allowing for precise positioning to any direction.

rise.risset explores the themes of ambiguity in localisation, sound source, and trajectories. The piece positions the four *speaker.motion* units in a square, enclosing the audience. The work is structured in three sections. The opening section explores the ambiguity in the cause of the sound. The ensemble creates mechanical noises as the speakers rotates and tilts. The piece starts with just these noises, using sharp twitches in rotation to create a rhythmic pattern. The noise itself is not loud, but audible, and as this section progresses, recordings of these mechanical noises are reproduced by the speaker. This amplifies the magnitude of these mechanical noises and the ensembles physical presence. This rhythmic material then leads into the next section.

The rhythmic pattern created in the opening section becomes the focus in the next section, with sharp noise bursts reinforcing the mechanical twitches. These bursts slowly fade in, blurring their entry, and eventually become the foreground. This rhythmic pattern becomes the basis of a risset rhythm, creating an impossibly accelerating trajectory. The layers are distributed across the speakers. This creates another layer to the experience of the risset rhythm, as the main layer which is perceived as the focus of the rhythmic figure slowly shifts around the ensemble. As this section comes to a close, the speakers start to slowly rotate, creating moments where the risset rhythm disintegrates into its multiple streams, as the amplitudes of the noises bursts are filtered by the space. This also acts as transitional material, moving into the last section.

The final section of the piece balances the energetic accelerating risset ryhthms with perpetual downwards pitch trajectories. The *speaker.motion* units create clear pitches when rotated at a constant rate, which is correlated to the speed of rotation. The four units are rotated slowly at different speeds, with the pitched noises tuned to create a chord. These noises become the pitch material for the last section. This blurs the cause of the sound, similar to the opening section, where the audience is unsure whether the mechanical noise of the system is what is being heard, or the speaker producing the sound is. The four speakers start to produce descending Shepard tones, using the pitched noise as their base. As each unit is rotating at different speeds, the audience is bathed in multiple layers of descending trajectories, which swirl around the performance space. This heightens the motion of the Shepard tones, constantly shifting the attention of the audience around the room. This downwards descent eventual exhausts itself, slowly fading out, with the speakers stopping one by one.

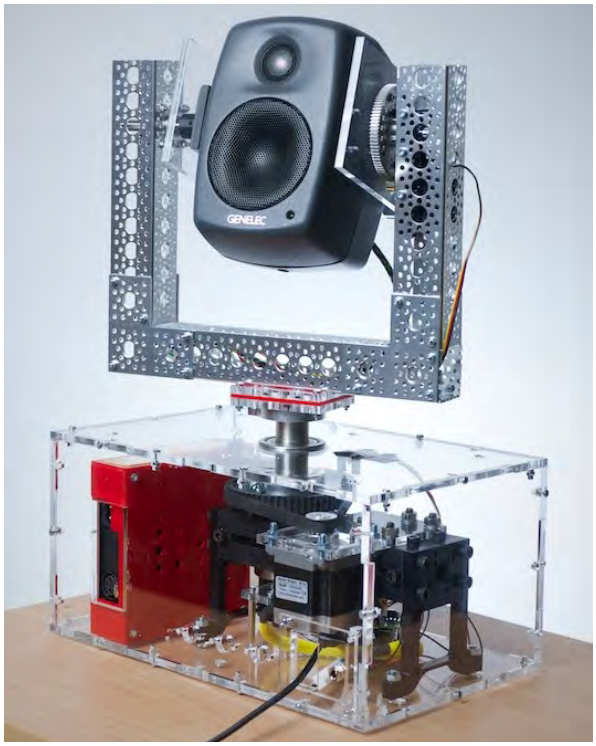


Figure 7. *rise.risset* – Blake Johnston for four *speaker.motion* units, Bridget Johnson's mechatronic speaker array.

6. FUTURE WORK

As evidenced in this paper, there is a broad range of auditory illusions and artefacts, which offer interesting insights to our perceptual processing, and potential as materials for artworks. The collection of works discussed and collected in this paper shows that there is a strong interest in using these themes in new sound works.

However, these materials are far from being exhausted. New tools like Alex Chechile's *The Eartone Toolbox* offers greater accessibility to composers and artists to explore Otoacoustic Emissions. Likewise, the ever developing field of new interfaces for musical expression shows advancements in the expressive control of space, through systems like Bridget Johnson's *speaker.motion.* The potential in these materials lie in their ability to create unique experiences for the audience that unveil and provoke questions about the subjectivity of their own perception of the world.

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